



FUNDING PROPOSAL

QUIET SHIP: SOLAR, BATTERY & EFFICIENT HOUSE POWER

Modern house power so the ship keeps her lights, security and communications on without running a diesel generator.

\$490,000

TOTAL REQUEST

Executive summary

The Foundation will install a modern, marine-rated house power system: a lithium-iron-phosphate battery bank, inverter/chargers, a deckhouse solar array, an LED lighting retrofit and remote monitoring. The system lets Comanche go “dead ship” at night without losing lighting, security cameras, bilge monitoring or internet — eliminating most generator hours, fuel cost, noise and emissions while at the pier or at anchor.

Statement of need

Comanche currently depends on a diesel deck generator for any electrical load when shore power is unavailable, which is most of the time at anchor. The ship goes dark at night: no security cameras, no reliable bilge alarms, no communications. Generator hours cost fuel, maintenance and volunteer attention, and a generator running overnight aboard a museum ship is a fire and carbon-monoxide risk. Reliable unattended power is also a prerequisite for the remote monitoring insurers increasingly expect of an unattended historic vessel.

Project description

Scope: (1) energy audit and system design by a marine electrical engineer to ABYC standards; (2) 60 kWh LiFePO4 house bank with battery-management system in a ventilated, fire-detected compartment; (3) two 12 kW inverter/chargers with distribution and transfer switching; (4) 12 kW solar array on deckhouse and awning frames with marine-grade mounts, plus a small wind turbine for winter; (5) LED retrofit of approximately 250 fixtures; (6) monitoring, telemetry, cameras and satellite/LTE internet; (7) ABYC-certified installation labor working with Foundation volunteers; (8) training and as-built documentation. The system is designed to be visible and interpretable — a teaching exhibit on how a 1944 ship meets a modern energy standard.

Goals & measurable objectives

- House power system installed, inspected and commissioned to ABYC standards.
- Generator run-hours reduced by at least 80% at anchor and pier.

- 24/7 lighting, cameras, bilge monitoring and internet without shore power.
- Lighting energy use reduced by ~70% through LED retrofit.
- Energy exhibit installed for visitors; volunteers trained on the system.

Timeline

Months 1–2	Energy audit, system design, equipment specification and bids.
Months 3–4	Equipment procurement; battery compartment preparation.
Months 5–7	Installation: batteries, inverters, distribution, solar, LED retrofit.
Month 8	Commissioning, monitoring setup, training.
Months 9–12	Performance logging; report to funder; exhibit installed.

Budget

Energy audit and ABYC system design (marine electrical engineer)	\$18,000
LiFePO4 house bank, 60 kWh, marine-rated with BMS	\$72,000
Inverter/chargers (2 × 12 kW), distribution, transfer switching	\$34,000
Solar array, 12 kW, marine mounts and awning frames	\$52,000
Wind turbine, 2 kW, mast and controller	\$9,000
LED lighting retrofit, ~250 fixtures	\$26,000
Battery compartment ventilation, fire detection and suppression	\$22,000
Monitoring, telemetry, security cameras, satellite/LTE internet	\$16,000
ABYC-certified installation labor, ~600 hours	\$78,000
Cabling, conduit, breakers, hardware	\$24,000
Training, as-built documentation, energy exhibit	\$8,000
Direct costs	\$359,000
Project management & administration (8%)	\$28,720
Contingency (25%)	\$96,930
Total project cost	\$484,650
Amount requested (rounded)	\$490,000

Budget lines are order-of-magnitude planning figures for a 133 ft / 295 GT steel vessel on Puget Sound, deliberately conservative so the funded scope can be delivered in full; they are replaced with vendor quotes at application. Unused contingency is returned or reallocated with funder approval. Partial funding is welcome and combinable with other sources.

In-kind and match. Foundation volunteer electricians and engineers (est. 800 hours); donated or discounted equipment sought from solar and battery manufacturers.

Evaluation

- Commissioning report and ABYC compliance letter.
- Generator hours and fuel use logged 12 months before vs. after.
- System uptime and monitoring records.
- Exhibit installed; visitor engagement recorded.

Funder fit

- Washington Clean Energy Fund / community energy programs
- Utility conservation and sponsorship programs
- Environmental and climate foundations
- Corporate sponsors (solar, battery, marine electrical)
- Community foundations

Sustainability

LiFePO4 banks carry 10+ year service life; solar has no fuel cost. Annual savings in generator fuel and maintenance fund the monitoring subscription and a component reserve.

Organization overview

Tug Comanche Historical Rescue Foundation (EIN 39-5018917) is a Washington nonprofit based in Auburn, Washington, listed on the IRS Publication 78 data list as eligible to receive tax-deductible charitable contributions (deductibility code PC). The Foundation preserves the historic vessel Comanche — laid down in 1944 as the U.S. Navy ocean tug ATA-202, later USS Wampanoag and Coast Guard cutter WMEC-202 — and operates her as a living platform for maritime heritage, hands-on education, youth mentorship, restoration-skills exposure, veteran and mariner storytelling, Puget Sound stewardship, and community service. Comanche is one of the last operational WWII museum ships in the world, still moving under her own power with an all-volunteer crew. The organization is governed by a volunteer board of directors and delivers its programs through volunteers, maritime-trades mentors and community partners.

Vessel history & significance

Comanche is a rare surviving example of a WWII-era U.S. Navy auxiliary ocean tug. Completed in 1944 and commissioned as ATA-202, she was named USS Wampanoag in 1948 and later served as the U.S. Coast Guard cutter WMEC-202 before becoming known as Comanche. Her current U.S. Coast Guard Certificate of Documentation (Official Number 647250; IMO 8991786) records a steel-hulled, mechanically propelled vessel of 133.4 feet registered length, 33.1-foot breadth, 17.7-foot depth and 295 gross tons (ITC), hailing port Seattle, Washington. She is propelled by two Cleveland Diesel 12-cylinder diesel-electric main engines driving a single screw, and holds a world record as the last ship still steered by deck-tube chain rudder control.

Legal & operating boundary. Comanche does not hold a U.S. Coast Guard Certificate of Inspection for commercial passenger service. No grant funds, fees or donations are or will be used for paid carriage, charters, towing, freight or any underway commercial service. Public programs take place alongside the pier; cruises occur only in partnership with other nonprofit and community organizations, never as a charter, and the Foundation retains control of the vessel at all times.

Contact: Wesley Weaver, Secretary — info@tug202.org — tug202.org · The web version of this proposal, with the latest figures, is at tug202.org/grants/solar-house-power.